

GlaBoost: A Multimodal Structured Framework for Glaucoma Risk Stratification

Cheng Huang, Zeyu Han, Weizheng Xie, Karanjit Kooner, Tsengdar Lee, Jui-Kai Wang, Jia Zhang[†]

Abstract—Early and accurate glaucoma detection is critical to prevent irreversible vision loss, yet existing AI methods often rely on unimodal inputs and lack interpretability. We present GlaBoost, a multimodal gradient boosting framework that unifies three complementary signals for glaucoma risk prediction: fundus image embeddings from a pretrained convolutional encoder, free-text neuroretinal rim assessments encoded by a transformer-based language model, and structured ophthalmic biomarkers. These modalities are fused into a single representation and classified by an enhanced XGBoost model. On two real-world annotated datasets, GlaBoost consistently outperforms unimodal and generic multimodal baselines. Feature importance analysis highlights the cup-to-disc ratio, rim thinning, and the ISNT rule as the dominant predictors, yielding clinically consistent and interpretable decisions. GlaBoost offers a transparent and scalable foundation for multimodal decision support in ophthalmology.

Index Terms—Ophthalmology AI, Glaucoma Risk Prediction, Extreme Gradient Boosting, Multimodal learning, Clinical Interpretability

I. INTRODUCTION

Glaucoma is a progressive optic neuropathy that ranks among the leading causes of irreversible blindness worldwide [20]–[22]. Early and accurate diagnosis is critical but remains challenging due to the disease’s complex presentation and its reliance on heterogeneous diagnostic evidence—fundus images [23], anatomical measurements [19], and expert clinical descriptions [18].

Recent AI-based approaches, particularly those built on convolutional neural networks (CNNs), have achieved strong performance in image-based glaucoma detection [19], [28], [30], [32], [35], [36]. However, these systems typically operate on a single visual modality and fail to incorporate the structured biomarkers and free-text annotations that ophthalmologists routinely rely on [18], [24]. This unimodal bias limits both diagnostic accuracy and clinical interpretability.

To bridge this gap, we propose GlaBoost, a multimodal learning framework that fuses fundus image embeddings, structured ophthalmic biomarkers, and transformer-encoded clinical narratives within an enhanced XGBoost classifier [4]. By explicitly modeling expert textual descriptions alongside

visual and numeric features, GlaBoost captures the semantic context that underlies clinical reasoning.

Our main contributions are summarized as follows:

- We propose the GlaBoost, a multimodal XGBoost-based framework that jointly leverages fundus images, structured anatomical biomarkers, and free-text clinical descriptions for glaucoma risk prediction.
- We design an effective pipeline for encoding free-text ophthalmic observations via a pretrained language model, and quantify their contribution to diagnostic performance through ablation studies.
- We demonstrate, across two real-world datasets and a comprehensive set of unimodal, multimodal, and large language model baselines, that multimodal fusion improves both predictive accuracy and feature-level interpretability in glaucoma diagnosis.

II. RELATED WORK

A. AI for Glaucoma

Most existing glaucoma AI systems are built around CNN-based architectures [21], [22], including U-Net variants [32], and have been applied to detection, segmentation, and reconstruction across diverse retinal imaging modalities [19], [23], [28], [29], [35]. Recent progress in large language models (LLMs) and vision-language models (VLMs) has further enabled the incorporation of textual evidence into medical AI [1], [2], [13], [25]. Nevertheless, current glaucoma AI remains predominantly image-driven [19], [28], [35], while clinical diagnosis depends on interpretable biomarkers such as intraocular pressure (IOP) and cup-to-disc ratio [6], [20], a mismatch that constrains both clinical adoption and trust.

B. Clinical Decision Making

In practice, ophthalmologists synthesize fundus imaging, patient history, and free-text descriptions of optic nerve head morphology to reach a diagnosis [6], [20]. Such textual descriptions encode semantic cues, e.g., rim pallor, notching, and bayoneting, that are difficult to recover from pixel-level features alone [18], [24]. Although recent explainable AI techniques offer interpretability through feature attribution or attention-based visualization [39], they are typically applied to a single modality and therefore cannot align with the inherently multimodal nature of clinical reasoning. GlaBoost is designed to close this gap by jointly modeling imaging, structured biomarkers, and expert narratives within a single interpretable framework.

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III. METHODOLOGY: GLABOOST

Data Input: let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the dataset, where each instance consists of multimodal features $\mathbf{x}_i$ and a binary label $y_i \in \{0, 1\}$ indicating the presence of glaucoma. We decompose the input feature vector $\mathbf{x}_i$ as:

$$\mathbf{x}_i = [\mathbf{x}_i^{\text{text}}; \mathbf{x}_i^{\text{struct}}; \mathbf{x}_i^{\text{human}}; \mathbf{x}_i^{\text{img}}] \quad (1)$$

where $\mathbf{x}_i^{\text{text}} \in \mathbb{R}^{d_t}$ is the textual embedding, $\mathbf{x}_i^{\text{struct}} \in \mathbb{R}^{d_s}$ represents structured clinical features, and $\mathbf{x}_i^{\text{img}} \in \mathbb{R}^{d_v}$ are visual features extracted from fundus images. $\mathbf{x}_i^{\text{human}}$ is the data based on human subjective judgment, not a label, and in certain datasets, it is not required and is considered optional.

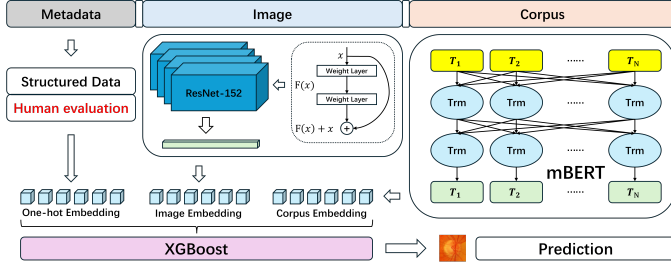


Fig. 1: The Architecture of GlaBoost

Based on this input of data, as shown in Fig. 1 GlaBoost have 4 stages: (1) **Structured Feature Extraction**; (2) **Image Feature Extraction**; (3) **Textual Feature Extraction** and (4) **Multimodal Fusion and Classification**.

A. Structured Feature Extraction

Each instance contains a clinical report field description where we extract two or one types of structured features:

1) *Machine-Readable Fundus Biomarker*: Within each report, the `fundus_features` field provides structured annotations of retinal morphology. We extract clinically established indicators, including optic disc size (small, normal, large) and the cup-to-disc ratio ($\in [0, 1]$). Categorical features are converted via one-hot encoding, and missing values are explicitly encoded as additional categories to ensure model compatibility, as formulated in Eq. 2:

$$\mathbf{x}_i^{\text{struct}} = \Phi(g(\text{description}_i)) \in \mathbb{R}^{d_s}, \quad (2)$$

where $g(\cdot)$ extracts raw fundus-related attributes, and $\Phi(\cdot)$ denotes a preprocessing function that performs one-hot encoding, normalization of continuous values, and explicit handling of missing entries. The resulting vector $\mathbf{x}_i^{\text{struct}}$ provides a fixed-dimensional representation of structured clinical information.

2) *Human-Evaluated Indicator (Optional)*: We extract two subjective diagnostic indicators provided by clinical annotators: `Glaucoma_risk_assessment` and `Confidence_level`. The glaucoma risk assessment is treated as a categorical judgment and encoded using one-hot vectors, while the confidence level is retained as a continuous score in $[0, 1]$. Both indicators are processed using the same feature extraction formulation as the structured data stage to obtain $\mathbf{x}_i^{\text{human}}$.

B. Image Feature Extraction

We apply a pretrained CNN, ResNet-152 [15] to extract image features from each retinal fundus photograph I_i :

$$\mathbf{x}_i^{\text{img}} = f_{\text{img}}(I_i; \theta_{\text{resnet}}) \in \mathbb{R}^{d_v} \quad (3)$$

The network is initialized with ImageNet-pretrained weights and extracts high-level image representations from the penultimate layer. All convolutional layers are frozen to reduce overfitting, with only the final projection layers optionally fine-tuned. Input images are resized to 224×224 , normalized using ImageNet statistics, and optionally augmented during training with random horizontal flips and color jitter.

C. Textual Feature Extraction

Each clinical report contains unstructured narrative fields such as some description, which often include nuanced spatial and qualitative assessments of optic disc morphology. To encode this textual modality, we use a Transformer-based language encoder f_{text} , mBERT [9]:

$$\mathbf{x}_i^{\text{text}} = f_{\text{text}}(T_i) = \text{mean-pool}(\text{Transformer}(T_i)) \in \mathbb{R}^{d_t} \quad (4)$$

where T_i denotes the input text token sequence and mean-pooling aggregates contextualized token embeddings. The model is initialized with domain-general weights and remains frozen to avoid overfitting due to limited textual supervision. We apply the following preprocessing to T_i : (1) Lowercasing and punctuation normalization; (2) Truncation or padding to a maximum sequence length and (3) Removal of medically irrelevant boilerplate text (optional).

D. Multimodal Fusion and Classification

The final feature vector is constructed by concatenating all modalities:

$$\mathbf{z}_i = [\mathbf{x}_i^{\text{text}}; \mathbf{x}_i^{\text{struct}}; \mathbf{x}_i^{\text{human}}; \mathbf{x}_i^{\text{img}}] \in \mathbb{R}^d \quad (5)$$

We use an XGBoost classifier [4] f_θ to predict the label:

$$\hat{y}_i = f_\theta(\mathbf{z}_i) \quad (6)$$

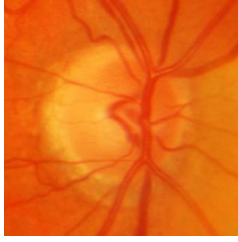
and optimize it with logistic loss:

$$\mathcal{L}(\theta) = - \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] + \Omega(f_\theta) \quad (7)$$

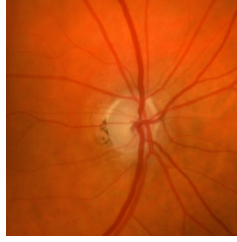
where $\Omega(f_\theta)$ is a regularization term controlling model complexity.

E. Dataset

Ethics and Data Governance. The experimental procedures involving human subjects described in this paper were approved by the Institutional Review Board (IRB) of the *University of Texas Southwestern Medical Center* (UTSW), in accordance with the Helsinki Declaration of 1975 (revised in 2000). As the study is retrospective, the IRB waived the requirement for written informed consent. All patient records and fundus images used in this work were fully de-identified prior to analysis, and no protected health information was accessible to the modeling pipeline.



(a) Public Dataset



(b) Private Dataset

Fig. 2: The Sample of Image from Public and Private Dataset

1) *Public Dataset*: TABLE I shows a representative sample from the public dataset, which is organized as JSON-formatted clinical reports paired with fundus images. During preprocessing, categorical fields such as `optic_disc_size` and continuous fields such as `cup_to_disc_ratio` are mapped to binary indicators (True/False) following established clinical thresholds for glaucoma diagnosis [6]. The free-text field `neuroretinal_rim` is retained as an unstructured description and processed through the textual encoder.

The two fields `glaucoma_risk_assessment` and `confidence_level` encode an annotator’s *preliminary risk tier* and *self-rated confidence*, respectively. We treat them as auxiliary clinician-derived inputs rather than ground-truth labels. A representative fundus image is shown in Fig. 2a. The dataset is publicly available¹.

TABLE I: Representative Sample from the Public Dataset

Field	Value
<code>optic_disc_size</code>	large
<code>cup_to_disc_ratio</code>	0.8
<code>isnt_rule_followed</code>	false
<code>rim_pallor</code>	true
<code>rim_color</code>	pale
<code>bayoneting</code>	true
<code>sharp_edge</code>	true
<code>laminar_dot_sign</code>	true
<code>notching</code>	true
<code>rim_thinning</code>	true
<code>additional_observations</code>	null
<code>neuroretinal_rim</code>	(free-text description)
<code>glaucoma_risk_assessment</code>	high risk
<code>confidence_level</code>	0.9

2) *Private Dataset*: TABLE II shows a representative sample from the private UTSW dataset, which is derived from clinical OCT reports. Each record contains 16 biomarkers organized into three anatomical categories, RNFL, ONH, and GCC, measured independently for OD and OS eyes, together with the corresponding IE.

Status Encoding Each biomarker is associated with a clinical status assigned by the OCT analyzer: *Within Normal*, *Borderline*, or *Outside Normal*. Borderline values lie within the normal reference range but approach the pathological boundary, and therefore carry diagnostic significance that is

distinct from both ends. We encode each biomarker status as an ordinal score $\{0, 0.5, 1\}$ corresponding to $\{\text{Outside Normal}, \text{Borderline}, \text{Within Normal}\}$, preserving the underlying clinical ordering. Unless a biomarker is explicitly marked as *Within Normal* in the report, *Borderline* cases are retained as a distinct third category rather than being collapsed into the *Normal* class.

Annotation All UTSW records are reviewed by two board-certified glaucoma specialists from the Department of Ophthalmology, UT Southwestern Medical Center, who provide the final diagnostic labels used as ground truth in this study. A representative fundus image is shown in Fig. 2b.

TABLE II: Sample from the UTSW Private Dataset

Category	Biomarker	Detail		
		OD	OS	IE (OD-OS)
RNFL	AR (μm)	97	91	6
	SR (μm)	94	92	2
	IR (μm)	99	89	10
	I-ER (S-I) (μm)	-5	3	-
ONH	A-O	0.28	0.51	-0.23
	V-O	0.46	0.79	-0.33
	H-O	0.62	0.74	-0.12
	RA (mm^2)	1.44	1.23	0.21
	DA (mm^2)	2.01	2.52	-0.51
	CVO (mm^2)	0.043	0.299	-0.256
GCC	A-G (μm)	85	83	2
	S-G (μm)	81	83	-2
	I-F (μm)	88	83	5
	I-EG (S-I) (μm)	-7	0	-
	FLV	0.87	0.47	0.40
	GLV	10.76	12.50	-1.74

Abbreviations: RNFL: retinal nerve fiber layer (AR: Average; SR: Superior; IR: Inferior; I-ER (S-I): intra-eye Superior-Inferior difference). ONH: optic nerve head (A-O / V-O / H-O: cup-to-disc area / volume / horizontal ratio; RA: rim area; DA: disc area; CVO: cup volume). GCC: ganglion cell complex (A-G: Average; S-G: Superior; I-F: Inferior; I-EG (S-I): intra-eye Superior-Inferior difference; FLV: focal loss volume; GLV: global loss volume). OD: right eye; OS: left eye; IE: inter-eye difference.

F. Experimental Setup

1) *Baseline Model*: we have chosen these following models as baseline models: RNN [11], Bi-RNN [34], LSTM [16], Bi-LSTM [16], [34], GRU [5], BERT [9], Transformer [39], Mamba [7], [12], RoBERTa [26], CNN [15], ViT [10], and ConViT [8]. Except for the last three CV models, CNN [15], ViT [10], and ConViT [8], all others are primarily designed for NLP tasks. We also choose the machine learning models: Linear Regression [14], Logistic Regression [17], Random Forest [3] and XGBoost [4]. For parameter optimization, we selected tool Optuna² as the optimization method.

TABLE III: Configuration for Fine-tuning with MLP Head

LLM	Version	DR	LR	BS	MSL	Epoch	GC
GPT	4O	10^{-1}	10^{-5}	8	512	5	1.0
	O3	10^{-1}	10^{-5}	8	512	5	1.0
LlaMA	3.1-405B	10^{-1}	10^{-5}	8	512	5	1.0
	3.1-70B	10^{-1}	10^{-5}	8	512	5	1.0
DeepSeek	R1	10^{-1}	10^{-5}	8	512	5	1.0
	V3	10^{-1}	10^{-5}	8	512	5	1.0

¹https://huggingface.co/datasets/AswanthCManoj/glaucoma_diagnosis_json_analysis

²<https://optuna.org/>

TABLE IV: Comparison of GlaBoost Against Machine Learning, NLP, Vision, Dual-Stream Multimodal, and Large Language Model Baselines on Public and Private Datasets

Model	Data Category			Judgment		Public Dataset			Data Category		Private Dataset			Cause
	Image	Words	Factor	Risk	Sure	ACC	PRE	F1	Image	Factor	ACC	PRE	F1	
Linear Regression	X	X	✓	✓	✓	45.72	51.03	48.26	X	✓	34.29	39.25	36.59	Yes
Logistic Regression	X	X	✓	✓	✓	64.37	67.29	65.83	X	✓	58.74	59.24	58.95	Yes
Random Forest	X	X	✓	✓	✓	78.93	81.77	80.30	X	✓	69.97	72.35	71.09	Yes
XGBoost	X	X	✓	✓	✓	98.22	97.86	98.04	X	✓	97.83	98.33	98.08	Yes
GlaBoost	X	X	✓	✓	✓	98.73	99.01	98.83	X	✓	99.21	98.89	99.10	Yes
RNN	X	X	✓	✓	✓	95.62	96.72	96.17	X	✓	95.50	95.38	95.44	Yes
Bi-RNN	X	X	✓	✓	✓	96.72	97.84	97.28	X	✓	97.58	97.04	97.31	Yes
LSTM	X	X	✓	✓	✓	97.44	98.34	97.89	X	✓	96.65	98.27	97.45	Yes
Bi-LSTM	X	X	✓	✓	✓	97.74	97.00	97.37	X	✓	96.73	96.79	96.76	Yes
GRU	X	X	✓	✓	✓	97.08	96.90	96.99	X	✓	97.76	97.78	97.77	Yes
BERT	X	X	✓	✓	✓	97.77	98.39	98.08	X	✓	98.72	98.02	98.37	Yes
Transformer	X	X	✓	✓	✓	97.86	98.90	98.38	X	✓	98.96	98.11	98.53	Yes
Mamba	X	X	✓	✓	✓	98.96	98.29	98.63	X	✓	98.68	97.57	98.12	Yes
RoBERTa	X	X	✓	✓	✓	98.10	98.57	98.33	X	✓	97.59	99.32	98.45	Yes
GlaBoost	X	X	✓	✓	✓	98.73	99.01	98.83	X	✓	99.21	98.89	99.10	Yes
CNN	✓	X	X	X	X	98.37	99.01	98.69	✓	X	98.82	98.24	98.52	No
ViT	✓	X	X	X	X	100.0	100.0	100.0	✓	X	100.0	100.0	100.0	No
ConViT	✓	X	X	X	X	100.0	100.0	100.0	✓	X	100.0	100.0	100.0	No
GlaBoost	✓	X	X	X	X	99.17	98.56	98.91	✓	X	98.37	98.52	98.41	No
CNN-RNN	✓	✓	✓	✓	✓	95.41	96.17	95.79	✓	✓	95.34	97.24	96.28	Yes
CNN-LSTM	✓	✓	✓	✓	✓	96.96	96.42	96.69	✓	✓	95.53	95.45	95.49	Yes
CNN-BERT	✓	✓	✓	✓	✓	97.02	96.61	96.81	✓	✓	97.43	96.49	96.96	Yes
CNN-Transformer	✓	✓	✓	✓	✓	96.56	96.81	96.68	✓	✓	97.15	95.11	96.12	Yes
ViT-Transformer	✓	✓	✓	✓	✓	95.73	96.90	96.31	✓	✓	96.91	96.71	96.81	Yes
ViT-Mamba	✓	✓	✓	✓	✓	96.83	97.38	97.10	✓	✓	97.38	95.64	96.50	Yes
ConViT-Transformer	✓	✓	✓	✓	✓	97.03	97.33	97.18	✓	✓	96.94	96.67	96.80	Yes
ConViT-Mamba	✓	✓	✓	✓	✓	96.97	96.48	96.72	✓	✓	97.41	96.88	97.14	Yes
GlaBoost	✓	✓	✓	✓	✓	98.36	98.52	98.71	✓	✓	98.96	99.03	98.91	Yes
GPT-4O	✓	✓	✓	✓	✓	96.13	96.02	97.65	✓	✓	96.34	96.57	96.43	Yes
GPT-O3	✓	✓	✓	✓	✓	97.82	94.74	94.35	✓	✓	95.78	96.00	95.89	Yes
DeepSeek-R1	✓	✓	✓	✓	✓	94.38	95.26	94.72	✓	✓	95.23	95.45	95.31	Yes
DeepSeek-V3	✓	✓	✓	✓	✓	96.27	94.4	95.85	✓	✓	94.81	95.13	94.95	Yes
LlaMA-3.1-405B	✓	✓	✓	✓	✓	96.33	95.49	96.76	✓	✓	94.18	94.42	94.32	Yes
LlaMA-3.1-70B	✓	✓	✓	✓	✓	97.94	96.87	94.94	✓	✓	94.49	94.80	94.61	Yes
GlaBoost	✓	✓	✓	✓	✓	98.36	98.52	98.71	✓	✓	98.96	99.03	98.91	Yes
GPT-4O	✓	✓	✓	✓	X	96.77	96.44	95.77	✓	X	94.81	94.37	94.52	Yes
GPT-O3	✓	✓	✓	✓	X	96.13	95.72	96.61	✓	X	95.14	95.67	<u>94.99</u>	Yes
DeepSeek-R1	✓	✓	✓	✓	X	95.57	97.29	96.65	✓	X	93.68	95.22	94.01	Yes
DeepSeek-V3	✓	✓	✓	✓	X	96.69	97.24	96.24	✓	X	94.75	93.93	94.32	Yes
LlaMA-3.1-405B	✓	✓	✓	✓	X	97.19	97.29	97.01	✓	X	93.52	94.13	93.78	Yes
LlaMA-3.1-70B	✓	✓	✓	✓	X	97.11	96.42	94.84	✓	X	95.89	95.14	95.32	Yes
GlaBoost	✓	✓	✓	✓	X	97.52	96.83	96.13	✓	X	95.42	94.76	94.93	Yes
GPT-4O	✓	✓	✓	✓	X	93.91	93.83	93.86	X	✓	94.78	94.90	94.83	Yes
GPT-O3	✓	✓	✓	✓	X	94.76	94.35	94.59	X	✓	94.12	94.21	94.17	Yes
DeepSeek-R1	✓	✓	✓	✓	X	94.15	94.08	94.11	X	✓	93.79	93.92	93.85	Yes
DeepSeek-V3	✓	✓	✓	✓	X	95.12	94.86	<u>95.01</u>	X	✓	93.47	93.59	93.53	Yes
LlaMA-3.1-405B	✓	✓	✓	✓	X	93.52	93.30	93.38	X	✓	93.05	93.12	93.09	Yes
LlaMA-3.1-70B	✓	✓	✓	✓	X	93.14	93.01	93.07	X	✓	92.82	92.91	92.86	Yes
GlaBoost	✓	✓	✓	✓	X	95.82	95.41	95.63	X	✓	95.21	95.38	95.29	Yes
GPT-4O	✓	✓	✓	✓	X	94.92	94.37	94.65	X	X	-	-	-	-
GPT-O3	✓	✓	✓	✓	X	95.11	94.50	94.80	X	X	-	-	-	-
DeepSeek-R1	✓	✓	✓	✓	X	<u>95.72</u>	<u>95.03</u>	<u>95.37</u>	X	X	-	-	-	-
DeepSeek-V3	✓	✓	✓	✓	X	94.48	93.70	94.09	X	X	-	-	-	-
LlaMA-3.1-405B	✓	✓	✓	✓	X	95.63	94.80	95.21	X	X	-	-	-	-
LlaMA-3.1-70B	✓	✓	✓	✓	X	94.65	93.94	94.29	X	X	-	-	-	-
GlaBoost	✓	✓	✓	✓	X	96.00	95.46	95.73	X	X	-	-	-	-

2) *Large Language Model*: we also use LLMs for evaluation, fine-tuning them with a classification head to align with the downstream task. They are divided into open source and closed source models. The open source model is LlaMA family [2], [37], [38] and DeepSeek family [13], [25], and the closed source model is GPT family [1], [31]. Regarding open-source LLM execution, DeepSeek family (R1, V3) [13], [25] were evaluated via their respective official APIs provided by the LLM developers. The LlaMA family (3.1-70B, 3.1-405B) [2], [37], [38] were accessed and run using the LlaMA-API platform. So do closed-source LLMs, like GPT family (4O, O3) [1], [31]. For the fine-tuning with the classification head (MLP, 2-layer and its size is 256 [33]), as shown in TABLE III it involves appending a task-specific classification head to the

LLM and jointly optimizing both components on the binary classification objective. By updating the model parameters during training, it enables the LLM to learn task-specific representations and decision boundaries. The Optimizer is the AdamW [27]. The Decay Strategy is the Cosine w/10% warm-up. Finally, the loss function is the Cross-Entropy³.

Note: Dropout Rate = DR; Learning Rate = LR; Batch Size = BS; Max Sequence Length = MSL; Gradient Clipping = GC.

G. Implementation

1) *Hyperparameter*: we implemented the multimodal classifier using XGBoost with standard settings, including a learning rate of 0.05, maximum tree depth of 6, 100 estimators,

³<https://docs.pytorch.org/docs/stable/generated/torch.nn.CrossEntropyLoss>

and `logloss` as the evaluation metric, with a fixed random seed for reproducibility. Image features were extracted using a ResNet-152 pretrained on ImageNet, with frozen convolutional layers and inputs resized to 224×224. Textual features were obtained using a transformer-based mBERT model kept frozen during training, with inputs normalized, truncated to 128 tokens, and pooled to form sentence embeddings.

2) *Hardware*: the hardware specifications for training and testing include 2 Tesla V100 GPUs (2 × 32GB), 64GB of RAM, 8 CPU cores per node, and a total of 6 nodes.

3) *Evaluation Metrics*: We evaluate classification performance using accuracy (ACC), precision (PRE), and F1-score (F1). Accuracy measures the proportion of correctly classified samples, precision reflects the reliability of positive predictions, and F1-score provides a balanced assessment under class imbalance. The best results are highlighted in bold, and the second-best are underlined (×100%).

IV. EXPERIMENTAL RESULT

A. Comparative Experiment

TABLE IV summarizes results across all baseline families. GlaBoost achieves the best accuracy, precision, and F1-score on both datasets. Vision-only models attain competitive numbers but cannot reason over structured biomarkers or clinical narratives, and are therefore marked “No” in the *Cause* column. Dual-stream multimodal encoders degrade as the input modality count grows, indicating that naïve feature concatenation is harder to optimize than tree-based fusion under limited data. Recent LLMs trail GlaBoost on most settings, suggesting that the diagnostic signal is dominated by a few structured biomarkers rather than long-form textual reasoning. Including clinician preliminary assessments raises performance for both LLM baselines and GlaBoost; we report results with and without such inputs side-by-side in TABLE IV to keep their contribution transparent rather than treating them as ground truth.

B. Ablation Experiment

1) *Public Dataset*: as shown in TABLE V, we evaluate the impact of different dataset components on model performance, both in isolation and in combination with other components. Using image-only inputs without any textual corpus, the model achieved performance comparable to that of multimodal approaches. However, it lacked clinical interpretability: while it could identify the presence of glaucoma, it failed to provide insights into the underlying biomarkers. Human evaluation factors, even when used in isolation, exert a measurable influence on the model’s performance. In scenarios where the model already achieves high precision, incorporating these human-derived factors often serves as the icing on the cake, further enhancing performance. Interestingly, when only images and vocabulary are provided, the model tends to “speak for itself by looking at the picture”, leveraging both visual features and text-derived embeddings to construct a representation of “glaucoma” for classification purposes.

TABLE V: Impact of Input Modalities of Public Dataset

Data Category			Judgment		Public Dataset			Cause
Image	Words	Factor	Risk	Sure	ACC	PRE	F1	
X	X	X	X	X	-	-	-	-
X	X	X	X	X	8.72	7.51	6.94	No
X	X	X	X	X	<u>6.38</u>	<u>5.87</u>	<u>6.01</u>	No
X	X	X	X	X	9.45	8.36	7.89	No
X	X	✓	X	X	97.69	97.84	97.55	Yes
X	X	✓	✓	X	97.98	98.22	97.81	Yes
X	X	✓	X	✓	<u>98.36</u>	<u>98.71</u>	<u>98.22</u>	Yes
X	X	✓	✓	✓	98.73	99.01	98.83	Yes
X	✓	X	X	X	46.27	49.13	47.12	No
X	✓	X	✓	X	47.85	50.72	48.66	No
X	✓	X	X	✓	48.94	51.03	49.71	No
X	✓	X	✓	✓	50.67	53.45	51.89	Yes
X	✓	✓	X	X	48.12	50.04	49.23	Yes
X	✓	✓	✓	X	49.03	51.67	50.11	Yes
X	✓	✓	X	✓	<u>50.45</u>	<u>52.31</u>	<u>51.18</u>	Yes
X	✓	✓	✓	✓	52.37	54.89	53.61	Yes
✓	X	X	X	X	99.17	98.56	98.91	No
✓	X	X	✓	X	98.97	98.42	98.73	Yes
✓	X	X	X	✓	99.01	98.44	98.80	No
✓	X	X	✓	✓	99.22	98.63	98.96	Yes
✓	X	✓	X	X	95.71	95.86	95.74	Yes
✓	X	✓	✓	X	95.89	96.12	95.91	Yes
✓	X	✓	X	✓	<u>96.21</u>	<u>96.18</u>	<u>96.10</u>	Yes
✓	X	✓	✓	✓	96.49	96.37	96.33	Yes
✓	✓	X	X	X	93.71	94.03	93.82	No
✓	✓	X	✓	X	<u>94.21</u>	<u>94.08</u>	<u>94.02</u>	Yes
✓	✓	X	X	✓	94.12	93.87	93.97	Yes
✓	✓	X	✓	✓	94.45	94.28	94.36	Yes
✓	✓	✓	X	X	96.00	95.46	95.73	Yes
✓	✓	✓	X	✓	95.82	95.41	95.63	Yes
✓	✓	✓	✓	X	<u>97.52</u>	<u>96.83</u>	<u>96.13</u>	Yes
✓	✓	✓	✓	✓	98.36	98.52	98.71	Yes

TABLE VI: Impact of Input Modalities of Private Dataset

Data Category			Private Dataset			Cause
Image	Factor	Borderline	ACC	PRE	F1	
X	X	X	-	-	-	-
X	✓	X	95.21	95.38	95.29	Yes
✓	X	X	<u>95.42</u>	94.76	94.93	No
✓	✓	X	98.96	99.03	98.91	Yes
X	X	✓	92.84	96.12	94.45	Yes
X	✓	✓	93.67	94.75	94.21	Yes
✓	X	✓	<u>94.91</u>	93.43	94.16	Yes
✓	✓	✓	97.38	97.01	97.21	Yes

2) *Private Dataset*: As shown in Tables II and VI, our dataset includes a *Borderline* category. Compared with the public dataset (12×2 variables), Table II contains approximately three times more parameter names, with substantially higher feature dimensionality (32×3 for single-eye and 48×3 for both-eye data). In the comparative experiments, *Borderline* cases are treated as normal by default, whereas in this analysis they are considered as a separate third class, resulting in an increase in variable dimensionality by factors of 4 or 2.7, respectively. As feature dimensionality increases, computational cost rises while classification accuracy degrades. Incorporating image features mitigates this effect by improving model fitting, leading to higher final accuracy.

3) *GlaBoost*: As shown in Table VII, we evaluate the individual components of GlaBoost through a series of ablation studies. Different CNN architectures are compared for image feature extraction, key mBERT hyperparameters are examined for textual feature modeling, and the influence of XGBoost parameters on biomarker-based classification is analyzed. Due to the relative simplicity of the classification task, CNN backbones such as ResNet-101 already achieve near-saturated accuracy, with deeper architectures yielding marginal gains. Textual representations extracted by mBERT improve

TABLE VII: Ablation Study of the GlaBoost

Model	Module / Function	Data Category			Judgment		Public Dataset			Data Category		Private Dataset			Cause
		Image	Words	Factor	Risk	Sure	ACC	PRE	F1	Image	Factor	ACC	PRE	F1	
CNN	+ ResNet-18	✓	✗	✗	✗	✗	91.20	89.10	90.14	✓	✗	90.80	91.10	90.95	No
	+ ResNet-34	✓	✗	✗	✗	✗	93.50	91.80	92.64	✓	✗	93.00	92.50	92.75	No
	+ ResNet-50	✓	✗	✗	✗	✗	94.80	93.60	94.19	✓	✗	94.50	94.10	94.30	No
	+ ResNet-101	✓	✗	✗	✗	✗	100.0	99.37	99.68	✓	✗	100.0	100.0	100.0	No
	+ ResNet-152	✓	✗	✗	✗	✗	100.0	100.0	100.0	✓	✗	100.0	100.0	100.0	No
	+ ResNet-200	✓	✗	✗	✗	✗	100.0	100.0	100.0	✓	✗	100.0	100.0	100.0	No
mBERT	+ Embedding	✗	✓	✓	✗	✗	87.79	88.31	88.05	✗	✓	87.90	87.46	87.68	No
	+ Attention	✗	✓	✓	✗	✗	91.26	89.83	90.54	✗	✓	90.77	91.02	90.89	No
	+ Dual Weight	✗	✓	✓	✗	✗	90.38	88.96	89.66	✗	✓	89.50	89.93	89.71	No
XGBoost	+ Default	✗	✗	✓	✓	✓	93.91	94.27	94.09	✗	✓	94.02	93.88	93.95	Yes
	+ Max_depth (10)	✗	✗	✓	✓	✓	94.62	95.13	94.87	✗	✓	94.91	94.45	94.68	Yes
	+ Subsample (0.8)	✗	✗	✓	✓	✓	94.35	93.89	94.12	✗	✓	94.41	94.18	94.29	Yes
	+ Random State	✗	✗	✓	✓	✓	95.02	94.71	94.86	✗	✓	95.13	94.96	95.04	Yes
	+ Fusion	✗	✗	✓	✓	✓	95.74	94.38	95.05	✗	✓	94.67	95.36	95.01	Yes
GlaBoost	+ ResNet-152	✓	✗	✗	✗	✗	94.88	95.36	95.11	✓	✗	95.22	94.95	95.08	No
	+ mBERT	✓	✓	✓	✗	✗	93.42	92.85	93.13	✓	✓	93.05	93.28	93.17	No
	+ XGBoost	✓	✓	✓	✓	✓	94.36	93.74	94.05	✓	✓	93.89	94.22	94.05	Yes
	+ All	✓	✓	✓	✓	✓	98.36	98.52	98.71	✓	✓	98.96	99.03	98.91	Yes

TABLE VIII: Statistics of Impact Factor (Top 5)

Public Dataset			Private Dataset		
Feature	Importance	Value	Feature	Importance	Value
isnt_rule_followed	0.5180	True	Superior RNFL (SR)	0.3424	91 - 96
cup_to_disc_ratio	0.2140	False	Cup/Disc H. Ratio (H-O)	0.3109	0.69 - 0.72
rim_thinning	0.2020	True	Average GCC (A-G)	0.1723	81 - 85
glaucoma_risk_assessment	0.0088	very healthy	Cup Volume (CVO)	0.1216	0.095 - 0.127
rim (embedding)	0.0132	thin (word)	FLV	0.0128	0.61 - 0.78

with parameter tuning, leading to consistent performance enhancements. For XGBoost, increasing model depth and varying random seeds improve both accuracy and robustness as feature dimensionality grows. Finally, the selected components are integrated into the full GlaBoost framework and jointly optimized, resulting in the best overall performance.

C. Analysis

As shown in Table VIII, feature importance analysis highlights the cup_to_disc_ratio and related attributes, including rim thinning and cup volume, as the most influential factors for glaucoma diagnosis across datasets. This is clinically well established, as an increased cup-to-disc (C/D) ratio reflects optic nerve damage due to retinal ganglion cell loss, with values above 0.6 or marked inter-eye asymmetry (> 0.2) raising strong suspicion of glaucoma [6], [20]. Similar diagnostic relevance is observed for the superior SRNFL. In addition, the included glaucoma risk assessment from the public dataset reflects human judgment that extends beyond a limited set of structured variables and implicitly incorporates visual cues not captured by existing annotations. This suggests that the non-image features in the public dataset are inherently incomplete.

V. CONCLUSION

We presented the GlaBoost, a multimodal gradient boosting framework that fuses fundus images, structured ophthalmic biomarkers, and free-text clinical descriptions for glaucoma risk stratification. Ablation analysis shows that the structured and textual modalities together recover most of the diagnostic signal even without any clinician-derived input, while feature importance analysis identifies clinically established predictors, the cup-to-disc ratio, rim thinning, and the ISNT rule, as the dominant contributors, in line with ophthalmologic practice.

Beyond the present setting, several directions remain open: adopting ophthalmology-specific foundation models for image encoding, replacing the general-purpose text encoder with biomedical language models, calibrating output probabilities for stratified clinical decisions, and validating GlaBoost across multi-site external cohorts.

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